

Astra: Histology-Derived Single-Cell Spatial Transcriptomics for Clinical Prediction

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Spatial transcriptomics ties gene expression to tissue architecture, but the assays remain expensive and sparse relative to hematoxylin-and-eosin (H&E) histology, which is abundant and already embedded in pathology workflows. We introduce **Astra**, a model that predicts spatial transcriptomics from H&E at single-cell resolution across diverse tissues. At inference Astra requires only H&E patch embeddings: it predicts the cellular layout within a patch and uses that inferred layout to assign per-cell gene expression, with no external cell segmentation or transcriptomic coordinates. Trained only on the HEST-1k Xenium corpus across breast, lung, bowel, and skin, Astra recovers structured, spatially coherent expression on held-out patients and on an out-of-distribution 10x breast-cancer cohort. On pathologist-annotated TCGA breast regions the same frozen model recovers tumor, lymphocyte, and stromal composition (per-slide correlation 0.855, 0.826, and 0.657), improving on every score reported by a breast-specific spatial-expression model evaluated on the same regions and folds. We then show that Astra’s H&E-derived expression carries clinical signal without task-specific fine-tuning of the image-to-expression model: it predicts trastuzumab response in HER2-positive breast cancer from tile-level predicted expression (AUROC 0.716), stratifies overall survival in ER-positive TCGA-BRCA (log-rank $p = 0.001$ between predicted risk quartiles), and pre-screens for microsatellite instability in colorectal cancer (AUROC 0.741). Together, these results show that histology-derived single-cell expression can support a variety of clinical applications, such as treatment-response prediction, prognosis, and molecular biomarker screening.

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Contents

1	Introduction	2
2	Model Overview	2
2.1	Astra recovers spatial expression from histology	2
2.2	Astra recovers pathologist-annotated cell composition	4
3	Architecture and data	6
4	Clinical Applications	11
4.1	Astra predicts trastuzumab response	11
4.2	Astra predicts survival risk in luminal breast cancer	13
4.3	Astra can help pre-screen for MSI-H	16
5	Discussion	17
6	Methods	18
6.1	Model architecture	18
6.2	Targets, loss, and optimization	19
6.3	Gene panel and training data	20
6.4	Whole-slide preprocessing	20
6.5	Inference: cell decoding and expression	20
6.6	Spatial-recovery evaluation	21

6.7 Cell-composition deconvolution	23
6.8 Downstream clinical models	23

1 Introduction

Spatial transcriptomics reads out gene expression while preserving where each cell sits, so it captures niches, tumor-immune boundaries, and local cell states that bulk or dissociated assays average away [1]. That spatial context is exactly what makes it valuable for understanding disease [2], yet the assays are costly, low-throughput, and sparse. Hematoxylin-and-eosin (H&E) histology is the opposite: it is cheap, ubiquitous, and already sits in every pathology archive. Learning to predict expression directly from H&E would turn that existing morphology into a molecular and spatial readout at cohort scale. That spatial readout matters clinically because many therapies act through cellular geometry rather than average composition: antibody-drug conjugates such as trastuzumab deruxtecan kill antigen-negative neighbors within a short diffusion radius (the bystander effect [3, 4]), so response in antigen-heterogeneous and HER2-low tumors depends on how positive and negative cells are interspersed [5]; likewise, checkpoint-blockade response tracks tumor-T-cell arrangement and organized tertiary lymphoid structures [6–8]. Each is a property of tissue geometry that survives only when a cell’s position is recorded alongside its expression.

Existing models establish that morphology carries transcriptomic signal, but their resolution or inference requirements limit how directly they can be used. Spot-level models such as ST-Net, HisToGene, and Hist2ST [9–11] learn from Visium-style targets, so each prediction averages over many cells and blurs the cell-to-cell variation that defines a niche. Single-cell methods such as GHIST and DeepCell, the Xenium-resolution variant of DeepSpot, improve resolution but still take the cellular layout as an input rather than inferring it: GHIST matches separate H&E and Xenium segmentations, while DeepCell crops H&E around supplied Xenium centroids [12, 13]. Either route ties training and inference to an external segmentation, and any spatial mismatch between the two assays injects label noise or discards usable training signal (Section 3).

Astra removes that dependency. At training, cells enter the model as queries defined by their centroids, while, in parallel, a density head learns where those centroids fall inside each patch. At inference, the density head proposes the layout from H&E alone and expression is queried at the inferred positions, without external segmentation or cross-assay matching. We first test whether this H&E-only pipeline recovers spatially coherent expression across tissues, and whether the same predictions reproduce the tumor, immune, and stromal composition that pathologists annotate on slides from a cohort the model never saw. We then hold Astra frozen and test three clinical applications in cohorts without measured spatial transcriptomics: trastuzumab response prediction in HER2-positive breast cancer, survival stratification in ER-positive breast cancer, and MSI-H pre-screening in colorectal cancer. These tasks test whether the predicted transcriptome provides useful clinical signal rather than only reproducing benchmark expression.

2 Model Overview

2.1 Astra recovers spatial expression from histology

We evaluate Astra with patient-level cross-validation on HEST-1k Xenium data restricted to breast, lung, bowel, and skin, withholding entire patients each fold. The primary metric is the per-gene

Spearman correlation between predicted and measured expression (Methods 6.6): cells with higher measured expression should rank higher in the prediction, which, because every cell has a coordinate, is equivalent to recovering where a gene is high or low on the slide. Genes vary widely in how much spatial structure they carry. Diffuse genes are nearly random across the tissue and hard to pin to morphology; genes that organize into compartments or neighborhoods are the ones Astra recovers (Figure 1a). Across each slide’s top-50 Moran’s I genes the median per-gene Spearman is 0.43 in breast, 0.42 in lung, 0.38 in bowel, and 0.32 in skin (0.40 pooled), a consistent level of recovery across four unrelated tissue types. Sweeping how many of those ranked genes are kept, recovery is highest for the most organized genes and falls as more diffuse genes are added (Figure 1b), so the top few dozen genes per slide carry the strongest predictions.

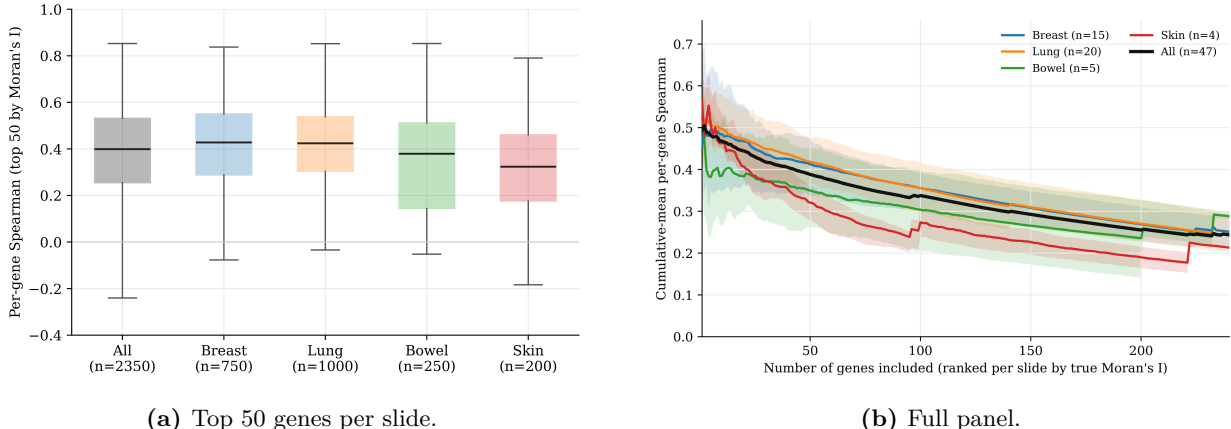


Figure 1: Held-out HEST-1k recovery with inferred layouts. Every slide ranks its own genes by descending true Moran’s I. (a) Distribution of per-gene Spearman over each slide’s 50 highest-ranked genes, pooled across the group’s slides, so one observation is one gene on one slide and every slide contributes 50 values; boxes give the interquartile range and whiskers the full extent. (b) The same quantity read cumulatively: each line is the mean across slides of the running average over the first n genes, with shaded bands the interquartile range across slides at that gene count. Skin ($n = 4$) and bowel ($n = 5$) contribute too few slides for their bands to be read as distribution estimates. Slides also carry different panel sizes, so each average in (b) is taken over the slides that still have genes remaining: all 47 contribute through 50 genes, 44 by 100, and 35 by 240, which is what produces the steps in the right-hand tail.

Aggregate correlations do not show what the predictions look like in tissue. We therefore examine one biologically informative gene from each tissue, always showing measured expression beside Astra’s prediction.

These maps test whether Astra places high-expression domains and compartment boundaries correctly rather than only matching a slide-level distribution, spanning four distinct morphology-expression relationships. GATA3 marks luminal mammary epithelium and estrogen-receptor signaling, a defining regulator of luminal breast-cancer identity [14]; COL17A1 anchors basal keratinocytes through hemidesmosomes and marks a distinct basal skin compartment [15]; FOXJ1 drives motile-cilia differentiation, localizing ciliated airway epithelium against surrounding lung [16]; and GREM1, a BMP antagonist, marks a cancer-promoting stromal program in colorectal tissue [17]. Astra recovers these domains with Spearman correlations of 0.53, 0.73, 0.71, and 0.62, respectively. These are qualitative spatial sanity checks chosen to cover distinct compartments (a nuclear luminal regulator, a basal adhesion protein, a ciliogenesis factor, and a secreted stromal antagonist), not the highest-correlation genes or evidence that they are privileged biomarkers; the held-out aggregate analyses (Figure 1) provide the quantitative assessment.

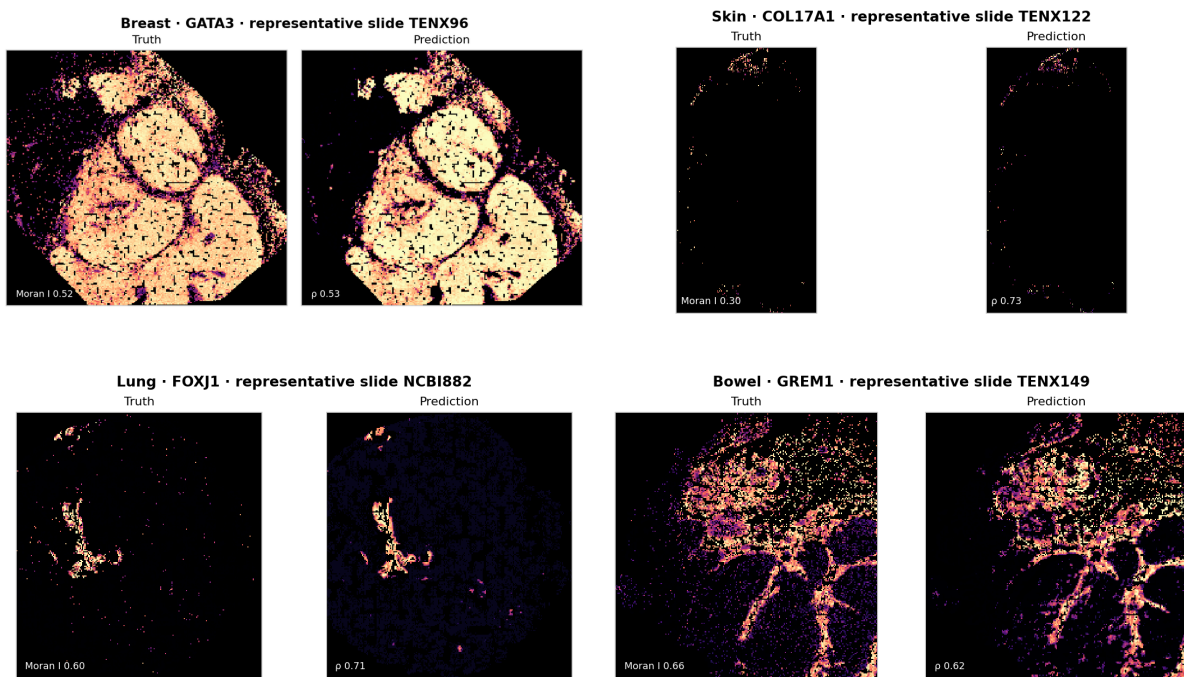


Figure 2: Measured (left within each panel) and Astra-predicted (right) expression for GATA3 in breast, COL17A1 in skin, FOXJ1 in lung, and GREM1 in bowel.

2.2 Astra recovers pathologist-annotated cell composition

The evaluations above compare Astra against measured transcripts. A different and stricter question is whether the predicted transcriptome encodes what a pathologist actually reads off the slide: how much of a field is tumor, immune infiltrate, and stroma. That composition is not an arbitrary target: tumor-infiltrating lymphocyte density is prognostic across breast-cancer subtypes [18] and has a standardized scoring protocol precisely because visual estimation is subjective [19]. Recovering it from H&E-derived expression therefore tests biological fidelity against a readout with established clinical meaning. We use PanopTILs [20], which fuses the BCSS region annotations [21] with nucleus-level classifications into 1,317 exhaustively annotated 1024×1024 regions of interest at $0.25 \mu\text{m}/\text{pixel}$ drawn from 118 TCGA breast-cancer slides. Every nucleus in the annotated field carries a manual class label, which gives each ROI a ground-truth fraction of tumor/epithelial, lymphocytic, and stromal cells. Nothing here is in distribution for Astra: the image-to-expression model is trained only on HEST-1k Xenium slides, sees no TCGA image, and is never shown a cell-type label.

Astra stays frozen. For each ROI we take the central 512×512 -pixel crop, which at $0.25 \mu\text{m}/\text{pixel}$ is exactly the $128 \mu\text{m}$ field in which PanopTILs concentrates its manual nucleus annotations, resample it to the model’s working resolution, and run the density and expression heads to decode cells and predict their expression. Averaging the decoded cells (median 155 per ROI) gives one 1,656-gene profile per ROI. A three-output multilayer perceptron then maps that profile to the three fractions (Methods 6.7), trained in the patient-grouped five-fold split released with Path2Space [22] so that no slide is split across folds, with genes and hyperparameters selected only inside each training fold. Following the same protocol, we score per-slide Pearson correlation between predicted and manual fractions and presence/absence AUROC, averaged over the 20 slides carrying at least 20 ROIs.

Predicted fractions track the manual counts closely for tumor and lymphocytes: per-slide PCC

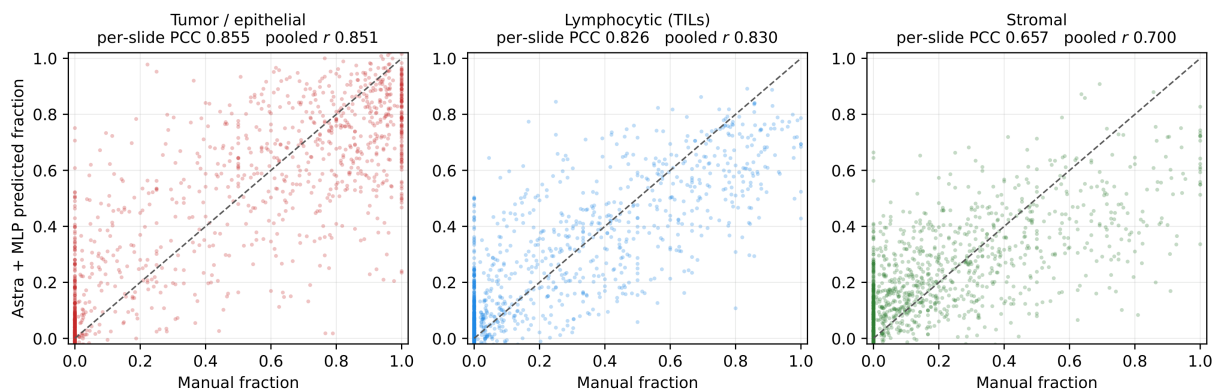


Figure 3: Held-out predicted versus manually annotated cell fractions across all 1,317 PanopTILs ROIs, with the identity line dashed. Panel titles give the per-slide correlation averaged over the 20 qualifying slides and the correlation pooled over all ROIs.

is 0.855 for tumor/epithelial and 0.826 for lymphocytic nuclei, with pooled correlations of 0.851 and 0.830 over all 1,317 ROIs (Figure 3). Stroma is harder (per-slide PCC 0.657, pooled 0.700) and far more variable between slides, ranging from 0.08 to 0.92, which is consistent with fibroblasts being the class whose morphology and expression overlap most with the surrounding tissue. Treated as a presence/absence call rather than a magnitude, the same predictions reach a mean per-slide AUROC of 0.915, 0.873, and 0.839 for tumor, lymphocytic, and stromal cells (Figure 4), so the model reliably answers whether a compartment is present in a field even where it misestimates how much of it there is.

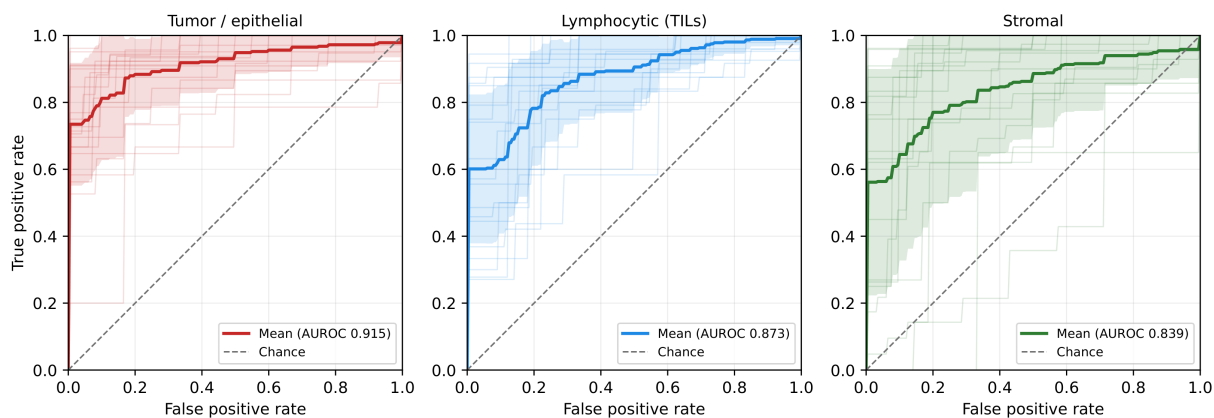


Figure 4: Presence/absence ROC curves per cell type. Thin lines are the 20 individual slides, the shaded band is ± 1 standard deviation, and the heavy line is the mean curve.

Because Path2Space reports this exact benchmark, the comparison isolates the expression model. The ROIs, the fold assignment, the gene-selection rule, and the regressor architecture and hyperparameters are all theirs; only the inferred expression fed into the probe differs. On that substitution Astra improves every one of the six reported scores (Figure 5): per-slide PCC rises from 0.793 to 0.855 for tumor, 0.789 to 0.826 for lymphocytes, and 0.592 to 0.657 for stroma, and presence AUROC from 0.876 to 0.915, 0.858 to 0.873, and 0.836 to 0.839. The margin is widest on the magnitude metric and narrowest on stromal detection, where both models are close to the same ceiling. The comparison also runs against the grain of the training data: Path2Space is trained on breast spatial transcriptomics, whereas Astra is trained across breast, lung, bowel, and skin and reaches the TCGA breast cohort as an out-of-cohort transfer.

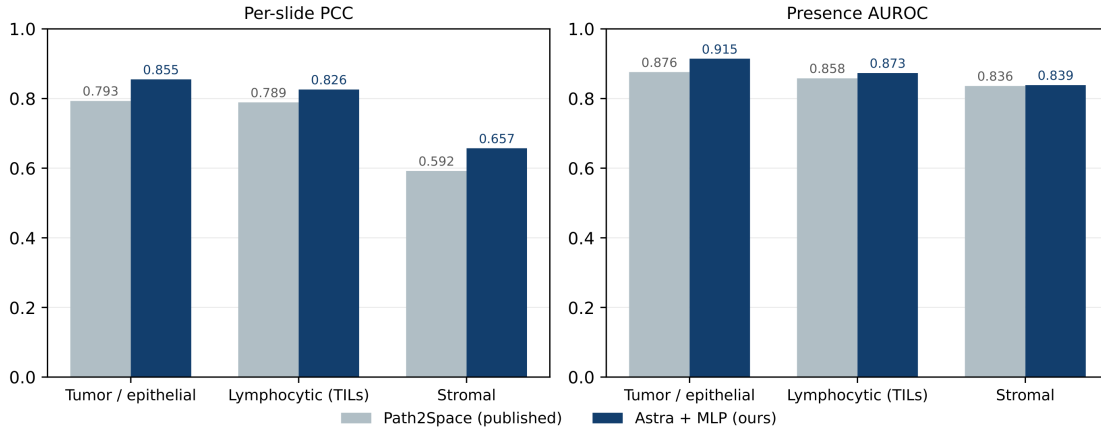


Figure 5: Astra against the published Path2Space scores on the same 1,317 ROIs, the same patient-grouped folds, and the same supervised probe.

Two limits frame this result. The probe is supervised, so it measures how much cell-composition information the predicted transcriptome carries rather than zero-shot deconvolution; we also applied reference-based deconvolution [23] directly to the same profiles and it performed substantially worse than the supervised probe, so we report the supervised route throughout. And the labels are annotations of morphology, not an orthogonal molecular assay, so agreement here establishes that Astra’s expression is consistent with expert reading of the tissue rather than independently validated against measured cell types. With those caveats, a frozen cross-tissue expression model recovers the compartment structure that TILs scoring depends on, on slides and a cohort it never saw, and does so more accurately than a breast-specific model built for the task.

3 Architecture and data

Astra uses a frozen H&E image encoder to convert each tissue patch into visual tokens, followed by a shared transformer trunk that integrates morphology across the patch (Methods 6.1). From this representation the model branches into three heads (Figure 6): a per-cell expression head, a detector head that predicts whether each target gene is non-zero in a given cell, and a density head that predicts where cells lie within the patch.

The HEST-1k Xenium slides do not share a single gene panel: the breast, lung, bowel, and skin assays each target a different set of transcripts. We reconcile these per-tissue Xenium panels into one harmonized vocabulary and train Astra to predict this fixed 1,656-gene output for every cell, regardless of which tissue the patch comes from. This single output space is what the downstream clinical analyses use: in Sections 4.1–4.3 each patient is run through the full 1,656-gene head, since those tasks operate per patient and never require a common gene set across Xenium slides. The per-slide recovery in Section 2.1 reports on far fewer genes: each slide is scored on its own top genes ranked by that slide’s Moran’s I, so the actual genes differ from slide to slide. To average a top- K statistic over all held-out slides on equal footing, every slide must be able to supply K genes, which caps K at the gene count of the smallest-panel slide, a few hundred genes. The number thus differs by application because of how the metric is aggregated, not because the model emits a different panel.

The central design decision is that cells enter as queries defined by their own centroids. Each cell’s relative position inside the patch is encoded with a Fourier embedding and used as a query

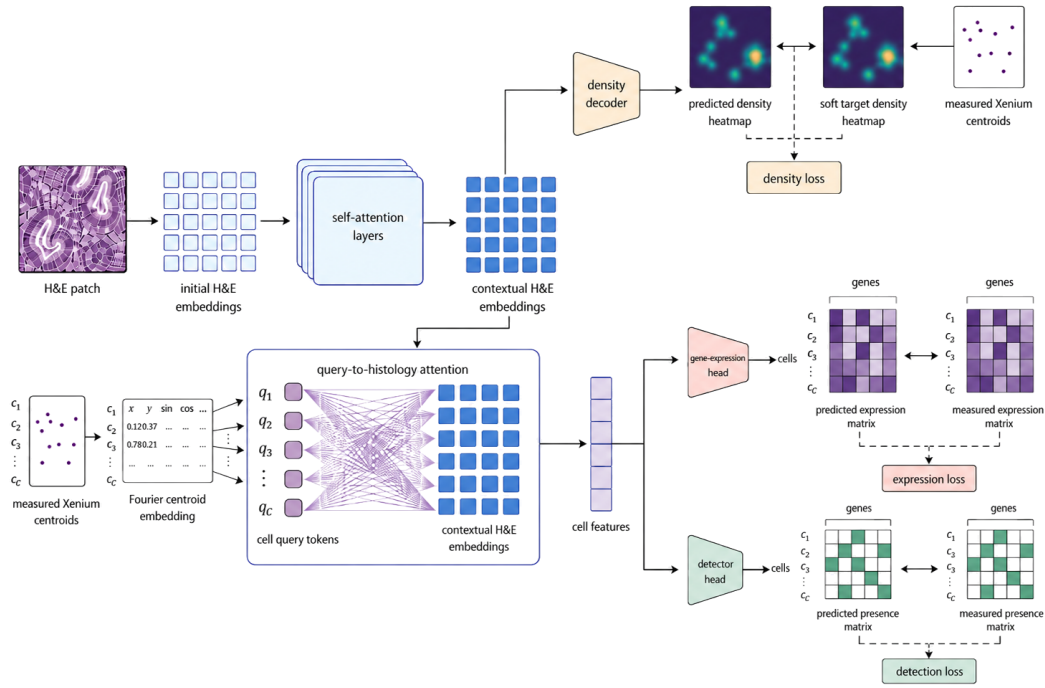


Figure 6: Astra’s shared visual trunk and position-conditioned density, presence, and per-cell expression heads.

that cross-attends to the patch’s image tokens, producing a feature for that specific cell. This avoids the object mismatch of methods that detect nuclei on H&E and then match them to a transcriptomics segmentation: Astra has no matching step, because the centroids themselves are the queries.

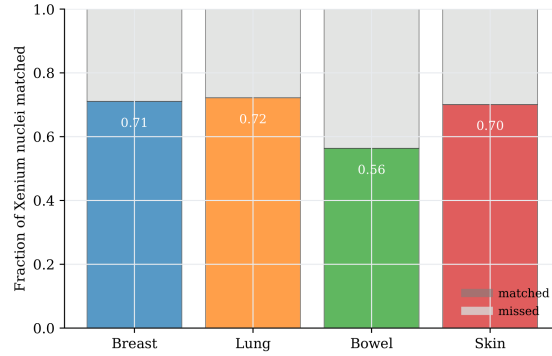


Figure 7: Xenium nuclei matched to H&E detections by tissue; grey, unmatched.

This is a design decision about how training supervision is used, not a benchmark against matching-based methods. H&E detections and Xenium nuclei are non-identical object sets: nearest-neighbor matching recovers 71% of Xenium cells in breast, 72% in lung, 56% in bowel, and 70% in skin, leaving 28–44% unmatched (Figure 7). Matched-pair supervision must therefore discard or misassign part of the measured transcriptome, whereas Astra anchors training queries to measured Xenium centroids and uses every measured cell, estimating their positions only at inference.

The density head is what makes inference H&E-only. During training, cell queries are placed at the true centroids while the density head learns the cellular layout from a unit-mass Gaussian

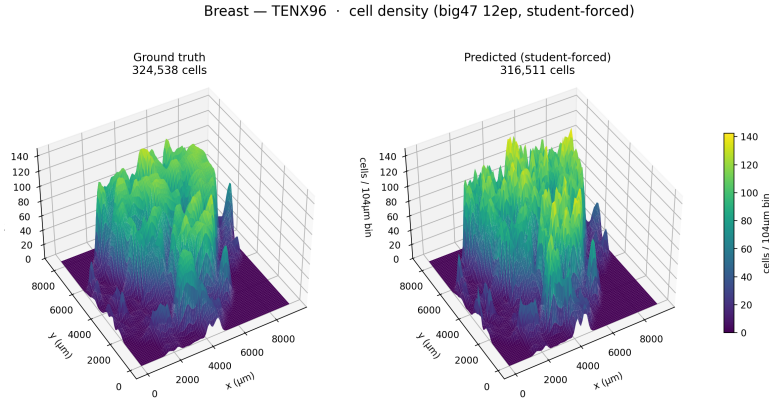


Figure 8: Measured and Astra-decoded density on held-out breast (104 μm bins; 324,538 vs. 316,511 cells).

target (Methods 6.2); at inference Astra proposes that layout and queries expression at the inferred positions. At inference the head emits a continuous density heatmap that must be turned into discrete cell positions. We decode it adaptively (Methods 6.5): because the density target is built from unit-mass Gaussians, integrating the map gives a cell count C , and comparing C against the number of resolvable local maxima says which decoder to trust. Where peaks account for at least 80% of C we take those non-maximum-suppression peaks directly; where the density saturates and peaks merge, so that fewer than 80% of the predicted cells appear as distinct maxima, we instead place all C cells as density-weighted centroids by a Lloyd/centroidal-Voronoi relaxation, which recovers the correct cell count in crowded regions where peak picking would undercount. The decoded layout closely tracks the measured one both in total count and in spatial distribution (Figure 8).

Because the decoded and measured cells are two independent point sets, we pair them for scoring by greedy nearest-neighbor matching (Methods 6.6): processed in increasing distance, each measured cell claims its closest unclaimed predicted cell within a distance gate set by the slide’s typical cell spacing, giving one-to-one pairs (unmatched cells count as misses or false positives), and per-gene Spearman correlations are computed over these matched pairs. All recovery results in this paper are scored in this fully H&E-only setting, using Astra’s decoded layout rather than measured centroids.

The detector head addresses a different failure mode. A continuous head trained with mean-squared error on $\log(1 + x)$ never outputs hard zeros, so absent genes emerge as small non-zero values and can fabricate infiltration for markers such as PTPRC [24]. A separate head predicts whether each gene is non-zero, and gating by that probability restores true absences. One fixed threshold, selected on held-out validation slides, is used throughout. The detector reaches AUROC 0.97 in lung, 0.94 in breast, 0.83 in bowel, and 0.80 in skin (pooled 0.94; Figure 9).

To sanity-check the training mixture and decide which tissues to keep, we trace each held-out query patch back to the training patches that most helped or hurt its prediction with influence functions. For a query patch q and a training candidate c the FastInf score is $I(c, q) = -(\nabla_{\theta} L_c)^{\top} \hat{G}_{\lambda}^{-1} (\nabla_{\theta} \ell_q)$, where ℓ_q is the all-genes MSE loss on the held-out patch and L_c the same loss on the candidate. The curvature term $\hat{G}_{\lambda}$ is a block-diagonal generalized Fisher matrix accumulated over a Fisher subsample of $N_F = 1,024$ training patches (64 batches of 16) as $\hat{G} \approx \frac{1}{N_F} \sum_i g_i g_i^{\top}$ per trainable matrix block (a diagonal Fisher for vector blocks), then *damped* by adding a multiple of the identity before inversion,

$$\hat{G}_{\lambda} = \hat{G} + \lambda I,$$

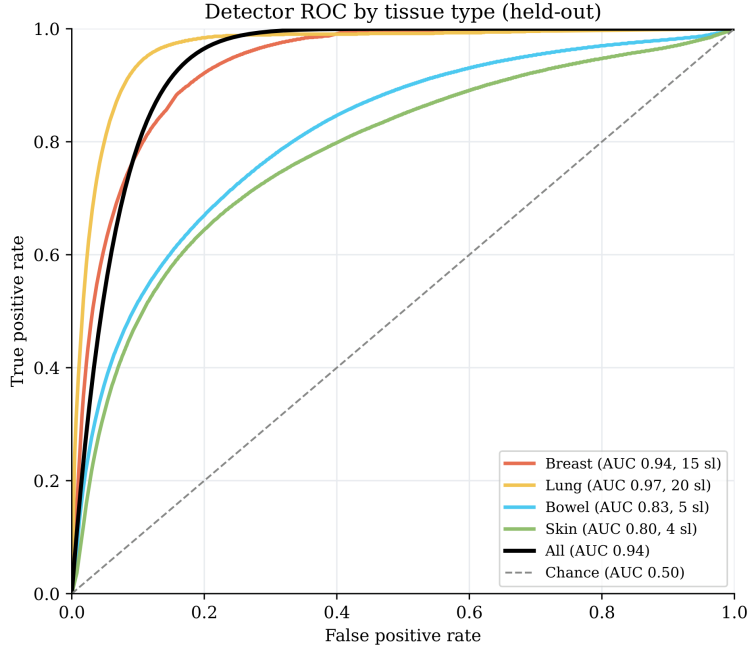


Figure 9: Held-out gene-presence detector ROC by tissue.

and inverted with a Schulz hyperpower iteration (HyperINF). The damping term λ is what makes the inverse well-conditioned: the empirical Fisher is low-rank and near-singular at this subsample size, so λ floors its spectrum and bounds $\|\hat{G}_\lambda^{-1}\|$, trading curvature fidelity for numerical stability. We use $\lambda = 0.01$, the smallest value in a $\lambda \in \{0.01, 0.05, 0.1\}$ sweep that still inverts stably, since larger damping drives $\hat{G}_\lambda^{-1} \rightarrow \lambda^{-1}I$ and degrades the score toward a plain gradient dot product. The score is signed: a negative value marks a beneficial (proponent) patch that lowers the query’s loss, a positive value a harmful (opponent) one. Scoring each query against the whole training set is intractable, so we narrow candidates in two stages. First a FastIF-style retrieval step picks *which* patches are eligible: for each query we take its $k = 10,000$ nearest training patches by L2 distance between frozen Gigapath class-token features, a purely visual-similarity filter. We then compute the influence score above for that shortlist and rank patches by the score itself, so “most influential” is decided by influence, not by feature distance. We report raw (non-cosine) scores at $\lambda = 0.01$ and $k = 10,000$ over 1,024 held-out queries.

The recovered relationships pass an obvious sanity check and surface a useful data decision. Beneficial influence is overwhelmingly same-tissue and morphologically matched: a query’s strongest proponents are training patches of the same tissue that share its morphology (same-tissue fraction 0.96 lung, 0.92 breast, 0.86 pancreas, 0.84 skin; Figure 10). The systematic exception is bowel, whose beneficial influencers are only 0.58 bowel and draw heavily on pale fibrous/stromal patches from pancreas (0.26) and breast (0.08), so influence tracks shared morphology, not the tissue label. That is what tells us to include the pancreas slides even though Astra is not intended for pancreas inference: pancreas is the single largest cross-tissue source of beneficial signal for bowel, a tissue we do report on. The qualitative panel shows the same pattern patch by patch, with each query retrieving same-tissue twins as its top proponents while bowel pulls in breast and pancreas stroma (Figure 11).

The magnitude of these scores is worth reading carefully, because ranks here are defined by influence rather than by feature distance (Figure 12). Lists are sorted by score descending, so rank 1 is a query’s *least* beneficial scored candidate. The median of those rank-1 scores is -3.8 (left),

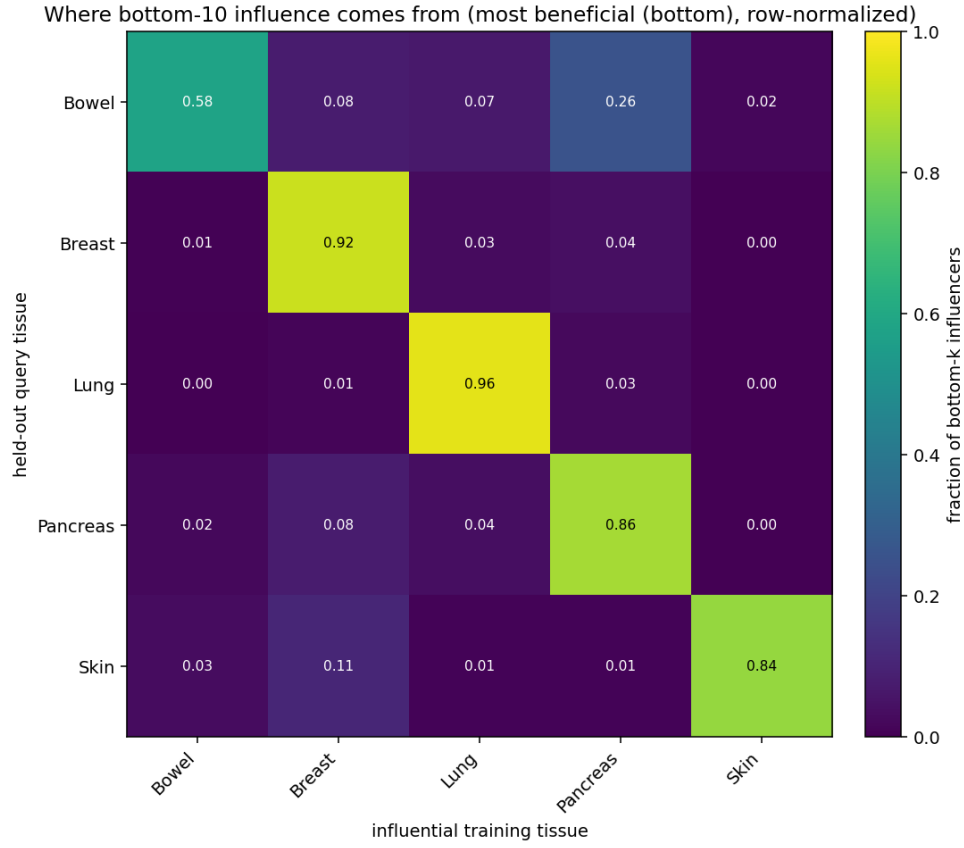


Figure 10: Where beneficial (bottom-10) influence comes from, row-normalized by held-out query tissue. Proponents are dominated by the same tissue, with bowel the exception (strong pancreas contribution).

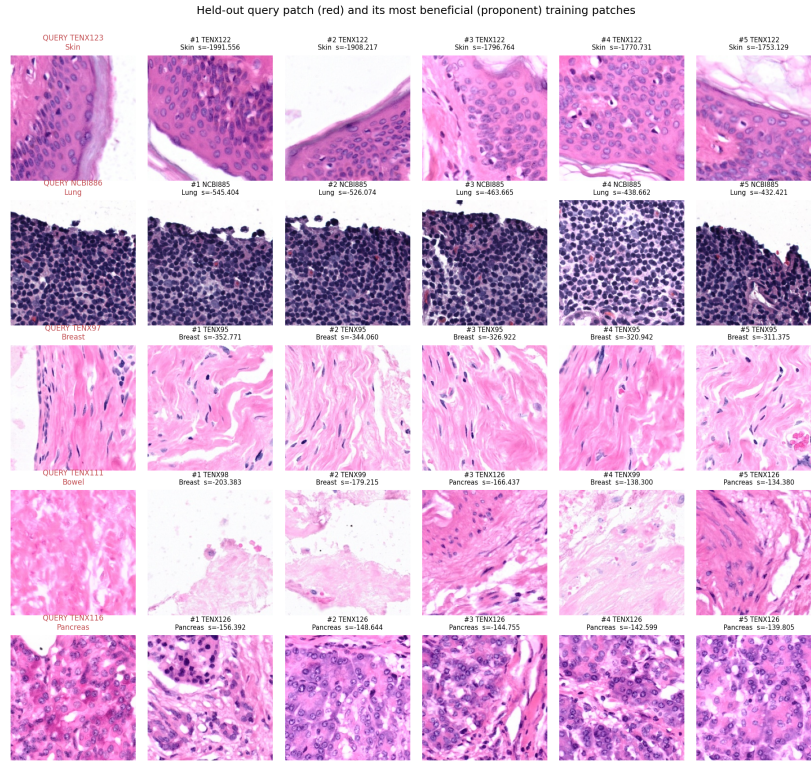


Figure 11: Held-out query patch (red, leftmost) and its top beneficial proponent training patches per row; same-tissue morphological twins dominate, while bowel retrieves breast and pancreas stroma.

already on the beneficial side of zero, which means that for a typical held-out query essentially the entire scored shortlist is signed beneficial and outright harmful candidates are the exception. The right panel shows the median and 10–90 percentile band of the score at each rank position once a query’s 10,000 candidates are sorted. It is therefore a sorted quantile profile of one distribution: its monotone descent is guaranteed by the sorting and carries no information about feature-space proximity or about how deep a kNN pool needs to reach. What it does report is the shape of that distribution. The bulk of candidates sit in a shallow band around the median, while the final decade of ranks steepens sharply, the median reaching about -22 and the 10th percentile falling below -60 . Strongly beneficial influence is thus concentrated in a small minority of patches rather than spread evenly over the shortlist, which is why the tissue attributions above are read from the bottom-10 proponents per query.

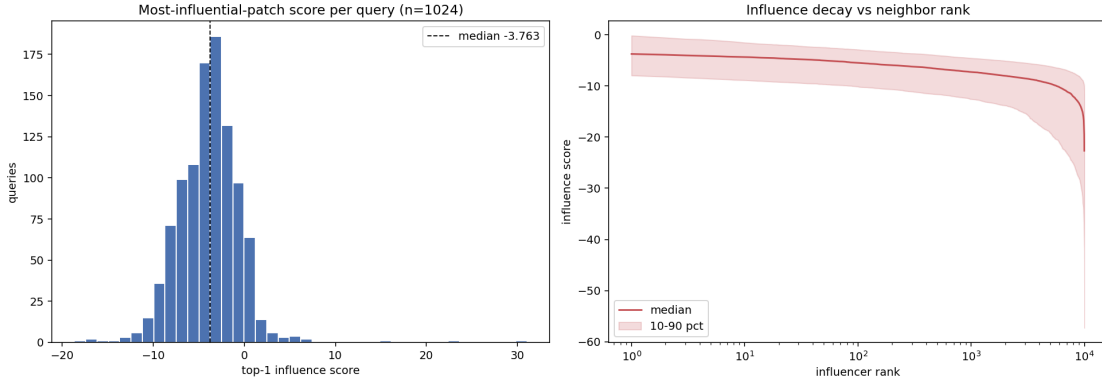


Figure 12: Influence score geometry over 1,024 held-out queries ($\lambda = 0.01$, $k = 10,000$, raw scores). Left: distribution of each query’s rank-1 (least beneficial) influence score, median -3.8 , already on the beneficial side of zero. Right: median and 10–90 percentile band of the score at each rank position after sorting each query’s candidates by influence. Ranks are defined by the score itself, so this is a sorted quantile profile of the score distribution and its monotone descent is a property of the sorting, not evidence about feature-space proximity; the sharp steepening in the last decade shows that strongly beneficial influence is confined to a small minority of candidates.

4 Clinical Applications

4.1 Astra predicts trastuzumab response

Having shown that Astra recovers spatial expression from histology, we now hold it frozen and ask whether that H&E-derived expression carries clinical signal in cohorts where spatial transcriptomics was never measured, beginning with treatment response. We use trastuzumab (anti-HER2) response in HER2-positive breast cancer on the Yale cohort (85 pre-treatment patients, 36 responders and 49 non-responders), labeled by pathologic response versus residual disease [25]. Astra stays frozen. Within each patient’s tumor ROIs (Methods 6.4) it detects cells and predicts their expression tile by tile, giving a stack of up to roughly 512 tiles across the 1,656-gene output panel per patient.

Rather than pooling every cell into a single slide-level vector, we keep the tile structure. Each tile is collapsed to a mean expression profile, and each gene is then summarized across the patient’s tiles three ways (Methods 6.8): the mean, the median, and the 90th percentile, a hotspot term weighting the most strongly expressing tiles. Each summary yields 1,656 features per patient. In each outer 5-fold split, the 17 test patients remain untouched. The feature budget is fixed at $K = 5$;

what the inner stratified 5-fold cross-validation over the 68-patient development set chooses is which summary family to use, by standardizing and ranking genes by response correlation on each inner training split, fitting an L2-regularized logistic regression on the top five, and scoring the held-out inner folds. We then refit the selected model on all 68 development patients before predicting the 17 test patients.

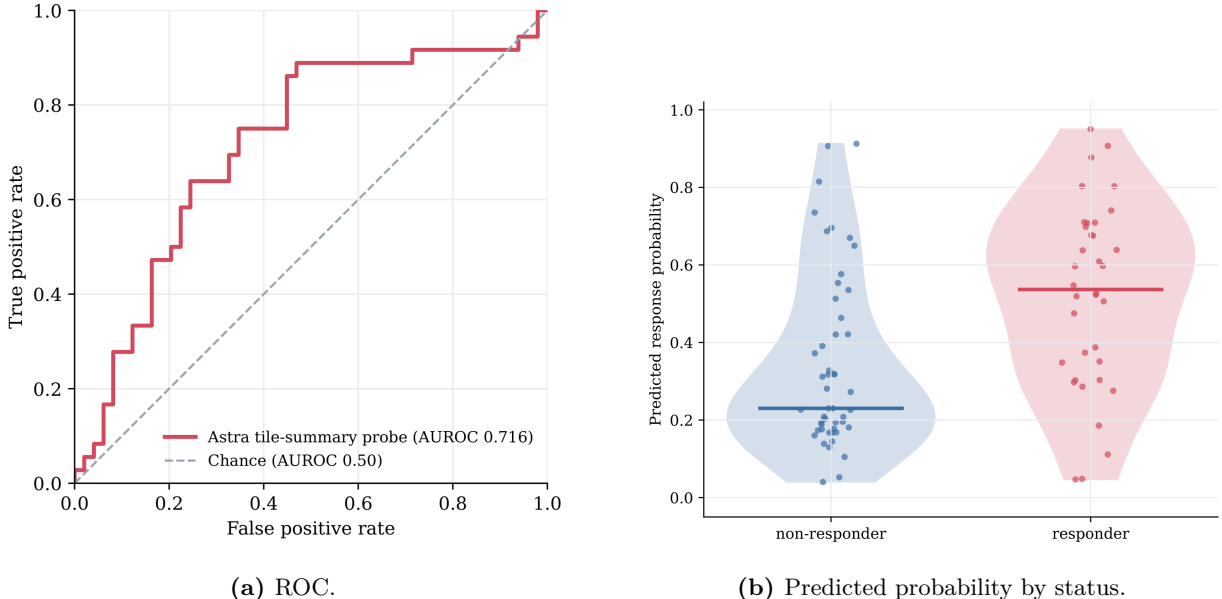


Figure 13: Trastuzumab response from Astra-derived tile-summary features ($n = 85$): ROC (AUROC 0.716) and out-of-fold probabilities.

Evaluated with repeated stratified 5-fold cross-validation (50 repeats) on averaged out-of-fold probabilities, this nested selection reaches AUROC 0.716 (95% CI 0.60–0.83 by 2000 $\times$ bootstrap) against a 0.42 response prevalence (Figure 13a). The inner cross-validation chose the hotspot (P90) summary in 230 of the 250 outer folds, the mean in 14 and the median in 6. The out-of-fold probabilities separate the two groups: responders receive a median predicted probability of 0.54 (IQR 0.34–0.70) against 0.23 (IQR 0.18–0.46) for non-responders, with group means of 0.52 and 0.34 (Figure 13b). The difference from prior work is one of supervision: on this same cohort, Farahmand et al. trained an end-to-end Inception-v3 CNN directly on the response label and reported roughly 0.80 slide-level AUROC [25], whereas Astra’s image-to-expression model was optimized only to predict expression, without task-specific fine-tuning on the response label; only the lightweight linear probe is trained on it.

Because the classifier is linear and its selection occurs entirely inside the training data, we can inspect which features remain stable without using test labels. Within the five-feature budget a small set of genes recurs: PCNA is retained by 217 outer models (87%), followed by MGST1 (132, 53%) and CYP2F1 (125, 50%), with the remaining genes each entering fewer than a third of models (Figure 14). The inner loop preferred the P90 summary in 230 of the 250 models, but we do not read that as evidence that heterogeneity carries the signal: across this cohort the three summary families are near-duplicates for these genes ($r \geq 0.91$), so the preference is marginal.

All three genes are elevated in non-responders, a direction the literature supports for one of them. MGST1 is a membrane-bound glutathione transferase that conjugates glutathione to electrophiles, reduces lipid hydroperoxides, and represses ferroptosis; its overexpression in MCF7 breast-cancer cells confers resistance to several cytostatic drugs and to oxidative stress [26], so enrichment in

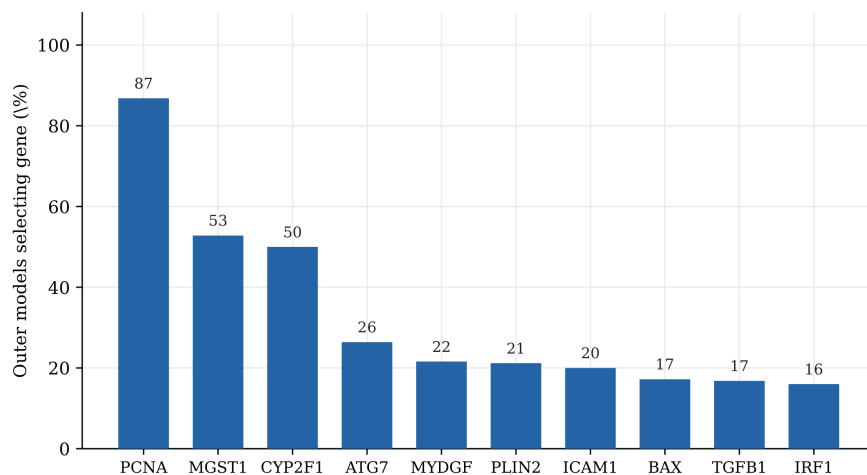


Figure 14: Top-10 genes by selection frequency: fraction of the 250 outer models (5 folds $\times$ 50 repeats) retaining each gene within its five-feature budget.

tumors that fail to clear is the expected direction for a detoxification program. PCNA is the sliding-clamp processivity factor for DNA polymerase; beyond its role as a proliferation marker it marks aggressive tumor features and poor prognosis in breast cancer [27], and it is named among the DNA-repair and damage-response genes differentially expressed in SKBR3 cells driven to trastuzumab resistance [28]. CYP2F1 is a lung-selective cytochrome P450 with no established role in breast tissue [29], and its recurrence most plausibly reflects structure in predicted expression rather than a resistance mechanism.

The directions above are univariate associations that do not adjust for one another.

Response here was predicted without any trastuzumab-specific training of the expression model: the same frozen Astra supplies the features and only a small linear probe is fit on the labels. This is the point of routing clinical prediction through predicted expression rather than training a bespoke supervised model per task. That expression is a shared substrate, so as the base model improves from aggregated spatial transcriptomics across more patients, its predictions become more accurate and every downstream readout built on them (response here, survival and MSI in the sections that follow) improves at once, with no new task-specific labels. The clinical results that follow lean on exactly this transfer.

4.2 Astra predicts survival risk in luminal breast cancer

Trastuzumab response used Astra’s expression to predict a treatment endpoint; we next ask whether the same frozen model carries prognostic signal for a different clinical outcome and cohort. Overall survival in luminal breast cancer is a natural test: molecular expression stratifies risk in ways morphology alone does not make obvious [30], yet measuring it at single-cell spatial resolution is rarely feasible at cohort scale. On 772 ER-positive TCGA-BRCA patients with matched diagnostic H&E and overall survival (97 deaths, 675 censored; median follow-up $\sim$ 865 days), we retain the 1,000 most tissue-dense tiles per patient (Methods 6.4), run frozen Astra to predict a 1,656-gene profile at each decoded cell, and average those predictions into one pseudobulk vector (Methods 6.8). Repeated 5-fold cross-validation (50 repeats) ranks genes by survival association within each training fold and fits a multivariable Cox model on the top 5 genes, the same fixed budget used for the other two tasks, and only held-out patients are scored, yielding a pooled concordance index of 0.603 (95%

CI 0.53–0.67).

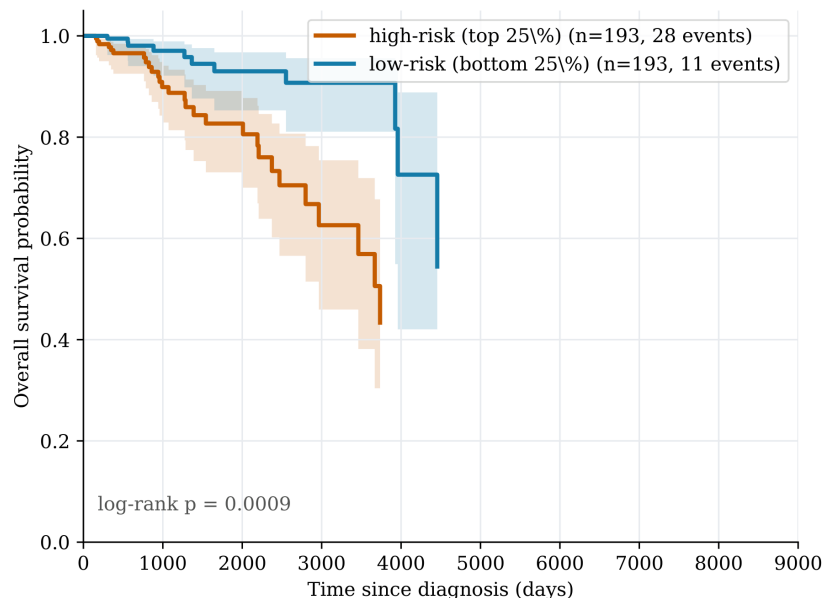


Figure 15: ER-positive TCGA-BRCA survival for the top and bottom cross-validated Astra risk quartiles ($n = 193$ each; two-sided log-rank $p = 0.001$).

The separation is sharpest at the extremes of predicted risk: comparing the top and bottom risk quartiles (193 patients each), the high-risk group has more than twice as many deaths as the low-risk group (28 vs. 11) and the survival curves diverge early and stay apart (Figure 15; two-sided log-rank $p = 0.001$). Splitting the full cohort at the median risk preserves the separation across all 772 patients (386 high vs. 386 low; 57 vs. 40 deaths; Figure 16; $p = 0.022$).

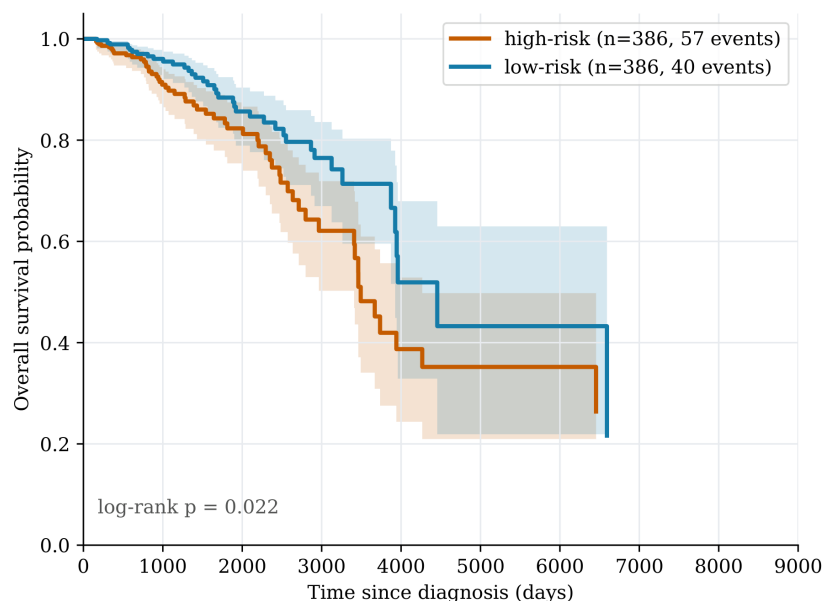


Figure 16: ER-positive TCGA-BRCA survival by cross-validated Astra risk at the median split ($n = 386$ high, 386 low; two-sided log-rank $p = 0.022$).

Because gene selection happens entirely inside the training folds, we can read off which genes

recur across the 250 outer models without touching held-out labels (Figure 17). Because the Cox coefficients are also recorded, each recurrent gene carries a direction. Two opposing programs emerge. Cytotoxic and humoral immune markers are consistently protective: KLRC1 (NKG2A; 58% of folds, mean coefficient -0.35), granulysin GNLY (22%, -0.34), and the immunoglobulin heavy chains IGHG4 (46%, -0.21) and IGHGP (32%, -0.26). Against them, SERPINB3 (88%, $+0.38$) and the myeloid calprotectin subunits S100A8 (58%, $+0.25$) and S100A9 (27%, $+0.27$) all carry adverse risk. Astra therefore recovers a coherent lymphocyte-versus-myeloid survival axis from H&E alone, with the sign of each gene consistent across folds.

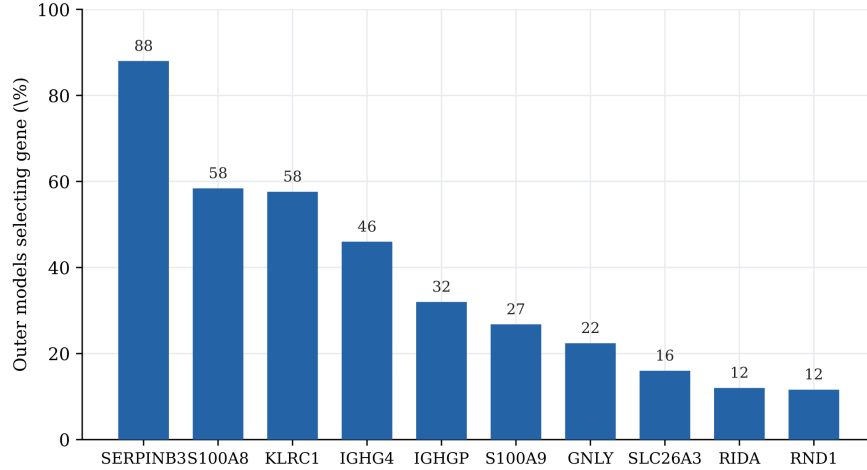


Figure 17: Top-10 genes by cross-validated selection frequency in the luminal survival model (fraction of the 250 outer Cox models retaining each gene).

Both directions match documented breast-cancer associations. Immunoglobulin expression reflecting infiltrating plasma cells is a validated favorable prognostic marker in breast cancer [31], which is the direction the probe assigns to IGHG4 and IGHGP, and dense tumor-infiltrating lymphocytes are a recognized favorable-prognosis axis [32]. Granulysin is a cytolytic effector released by cytotoxic T and NK cells, and GNLY-positive infiltrate is an independent predictor of better disease-free survival in breast cancer [33], matching the protective coefficient the probe assigns it. On the adverse side, the S100A8/S100A9 heterodimer marks a myeloid inflammatory program whose elevated expression in tumor cells and stroma is prognostic of poorer breast-cancer outcome [34, 35].

SERPINB3, the most frequently selected gene and the one with the largest adverse coefficient, carries the direction the literature reports. SERPINB3 (SCCA1) is elevated in high-grade, advanced-stage breast carcinoma and marks worse overall and recurrence-free survival [36], and in residual tumor after anthracycline-based neoadjuvant chemotherapy it is an independent prognostic biomarker of poor outcome (HR 2.1, 95% CI 1.2–3.8) [37].

KLRC1 admits no equivalent mechanistic reading. It encodes NKG2A, an inhibitory checkpoint receptor whose high expression on NK cells marks exhaustion and immune evasion, and which is an antibody target because blocking it restores cytotoxicity [38]. Its protective coefficient is therefore better attributed to pseudobulk KLRC1 transcript tracking the abundance of NK and cytotoxic T cells than to NKG2A signaling itself.

4.3 Astra can help pre-screen for MSI-H

Microsatellite instability (MSI-H) is among the most actionable biomarkers in oncology: it predicts response to immune-checkpoint inhibitors. Pembrolizumab is FDA-approved for MSI-H/dMMR solid tumors in a tissue-agnostic indication [39], and MSI-H also flags Lynch syndrome for genetic screening. Standard testing relies on PCR or immunohistochemistry for mismatch-repair proteins, so calling MSI from routine H&E would be a cheap pre-screen, and prior deep-learning work has shown histology carries this signal [40]. Our test is deliberately strict: the image-to-expression model receives no task-specific fine-tuning on MSI. It is trained only on HEST-1k spatial transcriptomics, with no TCGA data and no MSI labels anywhere in that model; only the linear probe on top sees the label. We run frozen Astra over TCGA colorectal slides (TCGA-COAD and TCGA-READ; 255 patients, 85 MSI-H), pool predicted per-cell expression into one pseudobulk profile per patient, and fit only a linear classifier on top under nested cross-validation (Methods 6.8; repeated stratified 5-fold outer CV, 50 repeats), keeping the top 5 AUC-ranked genes per training split and averaging over the mean and median tile summaries, scored against the genomic MANTIS label [41].

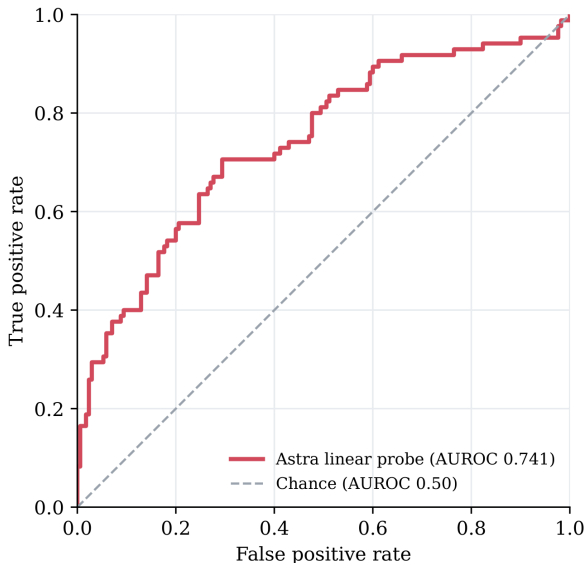


Figure 18: MSI-H detection from Astra-predicted pseudobulk expression in TCGA-COAD/READ (AUROC 0.741).

The linear probe reaches AUROC 0.741 at a cohort prevalence of 33% MSI-H (Figure 18). This is competitive with the within-cohort TCGA models trained end-to-end on MSI labels (patient-level AUC 0.77–0.84 [40]; our bootstrap 95% CI reaches 0.81). Astra reaches this without any of that task-specific machinery: it sees no MSI label or TCGA slide in training and fits only a named-gene linear probe, keeping an interpretability those end-to-end detectors lack. As argued for trastuzumab (Section 4.1), predicted expression is the shared substrate, so gains to the base model from aggregated spatial transcriptomics carry over to this task with no new MSI supervision.

Because the probe is linear and its genes are named, the prediction can be read directly. Hypermutated, neoantigen-rich MSI-H tumors recruit an inflamed immune infiltrate [42], and the two most stable features fit that picture. COTL1 enters essentially every probe (99%): it is an actin-binding protein of the ADF/cofilin family that is highly expressed in leukocytes and is required at the immune synapse, where it protects F-actin from cofilin-mediated disassembly and supports lamellipodial protrusion during T-cell activation [43]. CXCL5 enters 69% and is a CXCR2-ligand

neutrophil chemoattractant that is overexpressed in colorectal tumors and associated with advanced stage, liver metastasis, and poor prognosis [44]. Both are immune-compartment genes, which is the axis that separates MSI-H from MSS tissue.

Several of the remaining genes have colorectal literature, though none is a mismatch-repair marker. NMB (74%) is elevated in colorectal tumors relative to normal mucosa and is an independent predictor of poor survival, acting through USP21-stabilized NF- κ B signaling [45]. MARCKSL1 (64%) is an actin-regulatory protein of innate immune cells that is raised in the circulating extracellular vesicles of metastatic colorectal cancer patients and has been proposed to separate metastatic from non-metastatic disease [46]. HIST1H4C (64%) is a replication-dependent histone transcribed in S phase and so reads as a proliferation correlate. These mark tumor aggressiveness and immune content rather than mismatch repair.

The remaining features admit no such support. TSC22D1 (52%) and STRADB (47%) have no established colorectal or mismatch-repair association, and the recurrent selection of the melanocytic genes TYR (62%) and PMEL (11%) in colorectal tissue is more plausibly a panel artifact than a biological finding.

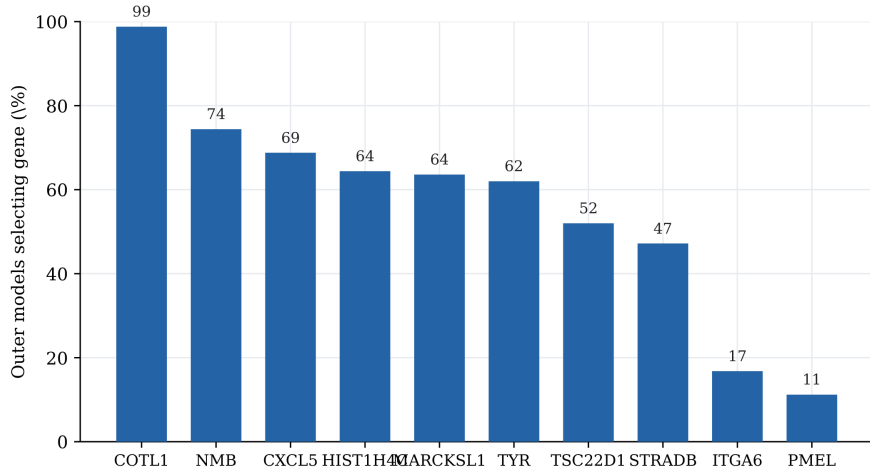


Figure 19: Top-10 genes by cross-validated selection frequency in the MSI-H probe. Each bar is the fraction of the 750 fitted probes (250 outer models $\times$ 3 regularization settings) in the gene’s dominant summary family that retain it.

A frozen, cross-domain expression model, without task-specific fine-tuning on MSI, therefore recovers an established clinical biomarker through an interpretable, named-gene route rather than an opaque classifier, and identifies the genes responsible. Two limits apply: these results are cross-validated within TCGA rather than externally validated, and the 33% MSI-H prevalence here is enriched relative to the roughly 15% observed clinically.

5 Discussion

Astra establishes a general architecture for learning single-cell molecular representations from routine histology. By jointly inferring cellular position and expression, it removes the requirement for a spatial assay or separately matched segmentation at inference. The same frozen representation transfers from expression recovery to pathologist-annotated cell composition, treatment-response prediction, survival stratification, and MSI-H pre-screening. Together, these findings suggest that

spatial-expression supervision can produce histologic representations with clinical utility across treatment response, prognosis, and biomarker screening. Prospective, multi-institutional validation is required before deployment.

This separation between a general image-to-expression model and a lightweight task-specific model is important. A single frozen model could be run retrospectively over existing pathology archives, after which small labeled cohorts could be used to test response, prognosis, or biomarker hypotheses without retraining Astra for every endpoint. Single-cell output also preserves regional heterogeneity that slide-level embeddings and pseudobulk measurements can erase. In practice, this could support molecular-test triage, identify tissue regions for confirmatory profiling, and help prioritize patients for biomarker-directed studies while retaining a spatial record of the predicted signal.

One particularly relevant use is clinical-trial enrichment, which we raise as a hypothesis to be tested prospectively rather than a demonstrated capability. FDA guidance defines predictive enrichment as selecting patients “more likely to respond to the drug treatment than other patients with the condition” and notes that this can increase effect size and permit a smaller study population [47]. This matters because aggregate trial results can obscure spatially defined differences in treatment response. In NeoTRIP, adding atezolizumab did not significantly improve pathologic complete response overall (48.6% vs. 44.4%, $p = 0.48$), but retrospective imaging mass cytometry of archived trial tissue found that cellular composition and spatial organization distinguished immunotherapy-sensitive from resistant tumors [48]. In NSABP B-41, the H&E-derived DeSTIL spatial immune signature identified a subgroup with markedly better event-free survival on trastuzumab than the combination arm (HR 0.09; interaction $p = 0.024$), while signature-negative patients showed no treatment difference [49]. Exploratory analysis of KATHERINE likewise found no T-DM1 benefit in the subgroup with focal HER2 expression below 30% (HR 1.21; focal-versus-nonfocal interaction $p < 0.05$), although the subgroup was small and the confidence interval wide (0.58–2.51) [50]. These retrospective results show how an aggregate trial effect can conceal treatment-relevant spatial organization. An inexpensive H&E-based pre-screen for a response-associated spatial program could prioritize patients for confirmatory testing or enrollment, potentially recovering efficacy signals that an unselected trial would dilute. Its near-term value is as a scalable screening layer that reduces the samples needing costly molecular assays, not as a determinant of eligibility on its own. More broadly, Astra shows how inferred single-cell expression can turn routine histology into a scalable substrate for biomarker discovery, trial enrichment, and clinically directed molecular profiling.

6 Methods

6.1 Model architecture

Astra maps an H&E patch and a set of cell positions to per-cell gene expression. Each patch is a 256×256 -pixel tile resampled to $0.5 \mu\text{m}/\text{pixel}$ (a $128 \mu\text{m}$ field of view), then bilinearly resized to 224×224 and ImageNet-normalized before the encoder. The image encoder is a frozen ProV-GigaPath ViT-G/16 [51] with embedding dimension 1536; we keep its full token grid, 197 tokens per patch (one class token and a 14×14 grid of patch tokens), rather than pooling to the class token.

The shared trunk is a four-layer pre-norm transformer over these 197 tokens with eight attention heads (head dimension 192). Each layer normalizes with root-mean-square norm (RM-

SNorm [52]), $\text{RMSNorm}(x) = x / \sqrt{\frac{1}{d} \sum_i x_i^2 + \epsilon} \odot \gamma$, and uses a gated SwiGLU feed-forward block [53], $\text{SwiGLU}(x) = W_{\text{down}}(\text{SiLU}(W_{\text{gate}}x) \odot W_{\text{up}}x)$ with inner width 4,096, in place of a GELU MLP. Token position is injected by two-dimensional rotary position embeddings (RoPE [54]) applied to the queries and keys: each head’s channels are split in half, one half rotated by the token’s grid column and the other by its row (base frequency $\theta = 100$; the class token is given the identity rotation). Before rotation, queries and keys are per-head RMSNorm-normalized (QK-norm) for training stability. Concretely, for query/key vector u at grid position with rotary tables $(\cos, \sin)$, $\text{RoPE}(u) = u \odot \cos + \text{rot}_{\text{half}}(u) \odot \sin$, and attention is the usual softmax over $q^\top k / \sqrt{192}$. A final RMSNorm is applied before the heads read from the trunk.

Cells enter as queries. A cell at pixel (x, y) inside the patch is expressed as a patch-relative coordinate $(dx, dy) \in [-1, 1]^2$ and embedded with a NeRF-style Fourier feature map [55] using eight frequency bands $\omega_k = \pi 2^k$ for $k = 0, \dots, 7$:

$$\phi(dx, dy) = [\sin(\omega_k dx), \cos(\omega_k dx), \sin(\omega_k dy), \cos(\omega_k dy)]_{k=0}^7 \in \mathbb{R}^{32},$$

which a linear layer projects to 1536 and an RMSNorm normalizes. This query cross-attends (eight heads) to that patch’s 197 image tokens, producing one 1536-dimensional feature per cell; the feature is RMSNorm’d and refined by a residual SwiGLU feed-forward block, $\text{SwiGLU}(x) = W_{\text{down}}(\text{SiLU}(W_{\text{gate}}x) \odot W_{\text{up}}x)$ with inner width 4,096. Because each cell is queried at its own centroid, there is no nucleus-to-transcript matching step.

Three linear heads read the per-cell feature. The expression head is a linear map to the 1,656-gene panel with no output activation. The gene-presence (detector) head is a parallel linear map to 1,656 logits predicting whether each gene is non-zero in that cell. The density head is separate: it reshapes the 14×14 patch tokens to a spatial grid, applies a 1×1 projection to 192 channels, upsamples three times ($2 \times$ bilinear, 3×3 convolution, GELU) to 112×112 , and applies a softplus to emit a non-negative density map.

6.2 Targets, loss, and optimization

Expression targets are library-size normalized to a fixed total count of 200 per cell and then $\log(1 + x)$ -transformed; all recovery metrics and downstream features are computed in this log-normalized space. Supervision is restricted to the genes each Xenium slide actually measures through a per-cell presence mask, so a slide never penalizes genes outside its panel. The training objective sums three terms,

$$\mathcal{L} = \lambda_{\text{expr}} \text{MSE}(\hat{y}, y) + \lambda_{\text{pres}} \text{BCE}_w(\hat{p}, \mathbf{1}[y > 0]) + \lambda_{\text{dens}} \text{MSE}_\Sigma(\hat{D}, D) + \lambda_{\text{cnt}} |\sum \hat{D} - N|,$$

with $\lambda_{\text{expr}} = 100$, $\lambda_{\text{pres}} = 1$ (positive class weight 4), $\lambda_{\text{dens}} = 1000$, and $\lambda_{\text{cnt}} = 0.1$. The expression term is mean-squared error in log-normalized space (the optional Pearson term carried by the `mse_pcc` loss is disabled). The density target D places a unit-mass Gaussian ($\sigma = 2$ density pixels) at each true centroid, scored by summed MSE, and a companion L_1 term matches the integrated density to the true cell count N . The model trains for 12 epochs with AdamW [56] (learning rate 7×10^{-4} , $\beta = (0.9, 0.95)$, weight decay 10^{-4} , gradient clip 1.0, batch size 48) under a 5% linear warmup followed by cosine decay to 10^{-6} . Patches are augmented with random flips, 90° rotations, light HED and color jitter; the encoder stays frozen throughout.

6.3 Gene panel and training data

Astra is trained on HEST-1k Xenium slides [57] across the target tissues. The per-tissue Xenium panels do not agree, so all measured genes are indexed into one universal axis and the 1,656-gene output is the union of genes measured anywhere in the corpus; each cell is supervised only on the genes its own slide measured. Patient-level cross-validation withholds entire slides per fold, and the 12 held-out slides used for the recovery analyses are disjoint from the 47 training slides.

6.4 Whole-slide preprocessing

For every downstream cohort, slides are tiled on a non-overlapping grid of 256-pixel tiles at $0.5 \mu\text{m}/\text{pixel}$, matching the training geometry. Each tile is stain-normalized by luminosity standardization: the tile is converted RGB to LAB, the 95th percentile p of the L (luminance) channel is taken, L is rescaled as $L' = \text{clip}(255 L/p, 0, 255)$ with the A and B channels unchanged, and the tile is converted back to RGB. This equalizes brightness per tile without a reference slide and leaves hue and saturation untouched.

Tiles are selected per task. For the TCGA-BRCA survival and MSI cohorts we rank tiles by tissue content and keep the top 1,000 per slide, where tissue is a mask of saturated, non-bright pixels (HSV saturation > 25 and value < 245) cleaned by 5×5 morphological opening and closing, and a tile’s score is the fraction of its pixels inside that mask. For the HER2 cohort we instead rasterize the pathologist tumor-ROI polygons and keep tiles whose ROI overlap is at least 0.5 and whose tissue fraction is at least 0.1, ranked by ROI coverage and capped at 512 tiles per slide.

6.5 Inference: cell decoding and expression

At inference Astra receives only H&E. For each tile the density head emits a non-negative map $\hat{D} \in \mathbb{R}_{\geq 0}^{H_d \times W_d}$ with $H_d = W_d = 112$ spanning the tile’s $128 \mu\text{m}$ field of view, so one density pixel is 2.29 source pixels ($1.14 \mu\text{m}$). Because the training target places a unit-mass Gaussian at every centroid, the integral of the map estimates the cell count,

$$\hat{N} = \sum_u \hat{D}_u, \quad C = \text{round}(\hat{N}),$$

and a pixel index $u = (r, c)$ maps to a patch-relative query coordinate by

$$\pi(u) = \left(\frac{2(c+\frac{1}{2})}{W_d} - 1, \frac{2(r+\frac{1}{2})}{H_d} - 1 \right) \in [-1, 1]^2.$$

Converting $\hat{D}$ into discrete centroids uses two decoders with complementary failure modes, selected per tile.

Peak decoding (non-maximum suppression). Let $\mathcal{N}_w(u)$ be the $w \times w$ box centered at u (zero-padded outside the map) and τ a peak-height floor. A pixel is a peak when it attains the maximum of its own neighborhood and clears the floor,

$$\mathcal{P} = \left\{ u : \hat{D}_u = \max_{v \in \mathcal{N}_w(u)} \hat{D}_v \text{ and } \hat{D}_u > \tau \right\},$$

and the decoded set is $\{\pi(u) : u \in \mathcal{P}\}$, optionally truncated to the C tallest peaks. We use $\tau = 0.005$ and $w = 3$. Two peaks at Chebyshev distance 1 would each have to dominate the other, so distinct

peaks are at least 2 density pixels ($\approx 2.3 \mu\text{m}$) apart, which sets the method’s resolution limit. Its cardinality $|\mathcal{P}|$ is emergent rather than prescribed: where neighboring Gaussians merge into a single plateau the peak count falls below C , so peak decoding undercounts in the most crowded regions.

Count-matched centroid decoding (Lloyd / centroidal Voronoi). The second decoder fixes the cardinality at C and solves only for position. Let $S = \{u : \hat{D}_u > \tau\}$ be the support, $p_u = \pi(u)$, and normalize the densities into a distribution on S ,

$$w_u = \hat{D}_u / \sum_{v \in S} \hat{D}_v, \quad \sum_{u \in S} w_u = 1.$$

We seek C sites $Z = \{z_1, \dots, z_C\}$ minimizing the density-weighted quantization energy

$$F(Z) = \sum_{u \in S} w_u \min_{1 \leq k \leq C} \|p_u - z_k\|_2^2,$$

which Lloyd’s algorithm attacks by alternating a Voronoi assignment with a weighted-centroid update. Writing $a(u) = \arg \min_k \|p_u - z_k\|_2$ and $R_k = \{u \in S : a(u) = k\}$, the update is

$$z_k \leftarrow \frac{\sum_{u \in R_k} w_u p_u}{\sum_{u \in R_k} w_u} \quad (R_k \neq \emptyset; \text{ empty regions hold } z_k \text{ fixed}).$$

Each half-step is non-increasing in F : the assignment minimizes F over a at fixed Z , and the weighted mean is the minimizer of weighted squared distance at fixed a . So F descends monotonically to a local minimum whose stationary point is the centroidal-Voronoi condition, every site sitting at the density-weighted centroid of the region it owns. Sites are initialized at support pixels drawn from $\text{Categorical}(w)$ (without replacement when $|S| \geq C$), which starts them inside the mass and keeps the iteration short. We stop at a tolerance of 10^{-5} on $\max_k \|\Delta z_k\|_\infty$ or a cap of 8 iterations, and return exactly C centroids. Unlike peak decoding this cannot undercount, but its points are centroids of mass rather than nuclei, so it is less accurate wherever peaks are genuinely resolvable.

Adaptive selection. The decoders are combined per tile by testing how much of the predicted count the peaks can account for. With $P = |\mathcal{P}|$ the uncapped peak count and a saturation fraction $s = 0.8$, peaks are trusted when $P \geq sC$ and the centroid decoder is used otherwise; $P \ll C$ is the signature of merged blobs. The ratio P/C is thus an estimate of the fraction of cells the map resolves as distinct maxima.

The decoded centroids are used directly as the cell queries of Section 6.1: each is Fourier-embedded and cross-attends to its own tile’s 197 image tokens, giving an $[n_{\text{cells}} \times 1656]$ matrix of predicted log-normalized expression per tile, and each is offset from its tile anchor into absolute slide coordinates so cells pool across tiles. The gene-presence probability, when used, is thresholded at a single value chosen on held-out validation slides to zero out predicted absences.

6.6 Spatial-recovery evaluation

Cell matching. The decoded cells and the measured Xenium cells are independent point sets of sizes M and N , so scoring first pairs them by greedy nearest-neighbor matching. Let $d_{ij} = \|c_i^{\text{true}} - c_j^{\text{pred}}\|_2$ be pairwise distances and τ the gate, set to the median nearest-neighbor spacing among measured cells. Candidate pairs with $d_{ij} \leq \tau$ are sorted ascending and consumed greedily, each cell used once: All per-gene correlations below are computed on the matched expression matrices $Y^{\text{true}}, \hat{Y} \in \mathbb{R}^{|\mathcal{M}| \times G}$ obtained by reindexing to $\mathcal{M}$.

Algorithm 1 Adaptive density decoding for one tile

Require: map $\hat{D}$, floor $\tau = 0.005$, window $w = 3$, saturation $s = 0.8$, iteration cap 8

- 1: $C \leftarrow \text{round}(\sum_u \hat{D}_u)$ $\triangleright$ count from the integrated density
- 2: $\mathcal{P} \leftarrow \{u : \hat{D}_u = \max_{v \in \mathcal{N}_w(u)} \hat{D}_v \text{ and } \hat{D}_u > \tau\}$
- 3: **if** $C > 0$ **and** $|\mathcal{P}| \geq sC$ **then**
- 4: **return** $\pi(u)$ for the $\min(C, |\mathcal{P}|)$ tallest $u \in \mathcal{P}$ $\triangleright$ peaks resolve the count
- 5: $S \leftarrow \{u : \hat{D}_u > \tau\}$; $w_u \leftarrow \hat{D}_u / \sum_{v \in S} \hat{D}_v$
- 6: $z_k \leftarrow \pi(u_k)$ with $u_k \sim \text{Categorical}(w)$, $k = 1, \dots, C$ $\triangleright$ peaks saturated
- 7: **for** $t = 1, \dots, 8$ **do**
- 8: $a(u) \leftarrow \arg \min_k \|\pi(u) - z_k\|_2$ for all $u \in S$
- 9: $z'_k \leftarrow (\sum_{u: a(u)=k} w_u \pi(u)) / (\sum_{u: a(u)=k} w_u)$ for non-empty regions
- 10: **if** $\max_k \|z'_k - z_k\|_\infty < 10^{-5}$ **then** $z \leftarrow z'$; **break**
- 11: $z \leftarrow z'$
- 12: **return** $z_1, \dots, z_C$

Algorithm 2 Greedy one-to-one cell matching

- 1: $\mathcal{C} \leftarrow \{(i, j) : d_{ij} \leq \tau\}$ sorted by increasing d_{ij}
- 2: $\mathcal{M} \leftarrow \emptyset$; $\text{used}_{\text{true}} \leftarrow \emptyset$; $\text{used}_{\text{pred}} \leftarrow \emptyset$
- 3: **for** $(i, j) \in \mathcal{C}$ **do**
- 4: **if** $i \notin \text{used}_{\text{true}}$ **and** $j \notin \text{used}_{\text{pred}}$ **then**
- 5: $\mathcal{M} \leftarrow \mathcal{M} \cup \{(i, j)\}$; mark i, j used
- 6: **return** matched pairs $\mathcal{M}$ $\triangleright$ unmatched cells are misses / false positives

Per-gene Spearman. Spearman's ρ is Pearson's r applied to ranks. For gene g , let $r(\cdot)$ assign average ranks (ties averaged) down the column of matched cells and $\bar{r}$ its mean:

$$a = r(\hat{Y}_{:,g}), \quad b = r(Y_{:,g}^{\text{true}}), \quad \rho_g = \frac{\sum_c (a_c - \bar{a})(b_c - \bar{b})}{\sqrt{\sum_c (a_c - \bar{a})^2} \sqrt{\sum_c (b_c - \bar{b})^2}}.$$

A gene with zero variance in either column is undefined and dropped from the aggregate; the headline number is the mean (or median) of ρ_g over kept genes. The out-of-distribution 10x slides instead report per-gene Pearson r on the matched values directly (no ranking).

Moran's I. Spatial structure per gene uses a k -nearest-neighbor graph over cell coordinates ($k = 6$), row-normalized to weights W with $\sum_j W_{ij} = 1$ (so $W_{ij} = 1/k$ for the k neighbors of i , else 0). For gene g with expression x , center it as $z = x - \bar{x}$. Since row-normalization makes $S_0 = \sum_{ij} W_{ij} = N$, the leading N/S_0 factor is 1 and Moran's I is the normalized quadratic form

$$I_g = \frac{z^\top W z}{z^\top z} = \frac{\sum_i z_i (Wz)_i}{\sum_i z_i^2}, \quad (Wz)_i = \frac{1}{k} \sum_{j \in \text{knn}(i)} z_j.$$

$(Wz)_i$ is cell i 's neighbor-averaged deviation, so the numerator rewards cells whose own deviation agrees in sign with their neighborhood: $I_g \rightarrow 1$ is smooth spatial clustering, $I_g \approx 0$ is spatially random.

Aggregation by spatial strength. Genes are ranked by I_g computed on *measured* expression, so selection never sees the prediction. Writing π for that descending order, the top-50 boxplot per

slide is $\{\rho_{\pi(1)}, \dots, \rho_{\pi(50)}\}$ (box = quartiles Q_1, Q_3 , line = median, Tukey whiskers to the farthest point within $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$), and the recovery-versus-gene-count curve is the running cumulative mean

$$S(n) = \frac{1}{n} \sum_{t=1}^n \rho_{\pi(t)},$$

averaged across slides within each tissue.

6.7 Cell-composition deconvolution

ROI representation. PanopTILs ROIs are 1024×1024 pixels at $0.25 \mu\text{m}/\text{pixel}$, with manual nucleus labels concentrated in the central $128 \mu\text{m}$ field. We crop that field (512×512 source pixels), resample it to the model’s 256×256 patch at $0.5 \mu\text{m}/\text{pixel}$, and apply the same luminosity normalization used for whole slides (Methods 6.4). Frozen Astra decodes cells from the density head and predicts expression at each decoded position (Methods 6.5); the ROI profile $x \in \mathbb{R}^G$, $G = 1,656$, is the mean over those cells.

Targets. Each ROI i carries manual fractions $y_i \in [0, 1]^3$ over tumor/epithelial, lymphocytic, and stromal nuclei. The three entries need not sum to one, since PanopTILs also labels classes outside this set (plasma cells, debris, and unlabeled nuclei), and they sum to 0.86 on average.

Probe. Within each training split we rank genes by the maximum univariate F -statistic across the three targets,

$$s_g = \max_{k \in \{1,2,3\}} F(x_{\cdot g}, y_{\cdot k}),$$

and keep the top 1,000. Features are min–max scaled to $[0, 1]$ on the training split and a single multi-output `MLPRegressor` with one 32-unit ReLU hidden layer predicts all three fractions jointly, fit with Adam (learning rate 10^{-3} , constant schedule, L2 penalty $\alpha = 0.8$, up to 1,000 iterations, early stopping on a 20% internal validation split with patience 50). These are the released Path2Space settings, adopted unchanged so that the benchmark isolates the expression model; hyperparameters are confirmed by an inner five-fold grid search inside each training split.

Evaluation. The five outer folds are Path2Space’s released patient-grouped assignment, so all ROIs from a slide fall in one fold and every ROI is scored by the model that never saw its slide. Out-of-fold predictions are pooled, and for each slide s with at least 20 ROIs we compute, per cell type k , the Pearson correlation between predicted and manual fractions and the AUROC of the predicted fraction against the binary presence indicator $\mathbf{1}[y_{ik} > 0]$. Both are averaged over the 20 qualifying slides (627 ROIs); the pooled correlations quoted alongside them use all 1,317 ROIs.

6.8 Downstream clinical models

The frozen Astra model is never fine-tuned on any clinical label; only a lightweight probe is trained per task. All fitting happens strictly inside each cross-validation training split, and all three tasks keep the same fixed budget of $K = 5$ ranked features so that they differ in endpoint rather than in capacity.

Patient representations. For survival and MSI we pseudobulk: with per-tile decoded expression $\hat{Y}^{(t)} \in \mathbb{R}^{n_t \times G}$ over retained tiles $t = 1, \dots, T$, the patient vector averages over *all* cells,

$$u = \frac{\sum_t \sum_{c=1}^{n_t} \hat{Y}_{c,:}^{(t)}}{\sum_t n_t} \in \mathbb{R}^G, \quad G = 1,656.$$

For HER2 we keep the tile structure instead of pooling across it. Each tile is collapsed to its mean profile $m^{(t)} = \frac{1}{n_t} \sum_c \hat{Y}_{c,:}^{(t)}$, transformed as $\tilde{m}^{(t)} = \log(1 + \max(m^{(t)}, 0))$, and each gene g is summarized across the T tiles in one of three ways,

$$\text{MEAN}_g = \frac{1}{T} \sum_t \tilde{m}_g^{(t)}, \quad \text{MED}_g = Q_{50}(\tilde{m}_g^{(\cdot)}), \quad \text{P90}_g = Q_{90}(\tilde{m}_g^{(\cdot)}),$$

each giving a $G = 1,656$ -dimensional feature vector per patient; only tiles with at least 25% ROI overlap contribute. The inner loop selects which family to use. Across patients the three are near-duplicates for the recurrently selected genes ($r \geq 0.91$ between families, and $r \geq 0.99$ between mean and median), so the family choice is a weak preference rather than evidence that any one of them isolates a distinct signal.

Survival (TCGA-BRCA, luminal). With right-censored overall survival (t_i, δ_i) we fit a Cox proportional-hazards model $h_i(t) = h_0(t) \exp(\beta^\top x_i)$ by maximizing the partial likelihood over the risk sets $R(t_i) = \{j : t_j \geq t_i\}$,

$$\ell(\beta) = \sum_{i:\delta_i=1} \left[\beta^\top x_i - \log \sum_{j \in R(t_i)} e^{\beta^\top x_j} \right].$$

Within each training fold we standardize genes, rank them by univariate association (each gene's single-covariate Cox score), and keep the top $K = 5$, the fixed budget shared by all three downstream tasks; the multivariable model is then fit on those genes. The risk score for a held-out patient is $\hat{\beta}^\top x$; concordance is Harrell's C , the fraction of comparable patient pairs whose predicted risk orders agree with survival:

$$C = \frac{\sum_{i:\delta_i=1} \sum_{j:t_j > t_i} \mathbf{1}[\hat{\beta}^\top x_i > \hat{\beta}^\top x_j]}{\sum_{i:\delta_i=1} \sum_{j:t_j > t_i} 1}.$$

This is repeated over 5-fold cross-validation (50 repeats); out-of-fold scores are pooled, and patients are compared by a two-sided log-rank test both at the median split and between the top and bottom risk quartiles.

Response and MSI (logistic probe). Both binary tasks fit an L2-regularized logistic regression on standardized features,

$$\min_{w,b} \frac{1}{2} \|w\|_2^2 + C \sum_i \log(1 + e^{-y_i (w^\top x_i + b)}), \quad \hat{p}_i = \sigma(w^\top x_i + b),$$

with $y_i \in \{-1, +1\}$. MSI (TCGA-COAD/READ) fits this on the pseudobulk vector against the genomic MANTIS label under a repeated stratified 5-fold outer loop (50 repeats). Within each training split genes are ranked by univariate AUC and the top $K = 5$ retained; because nothing further is selected, the model is averaged rather than chosen, ensembling the two tile summaries (mean and median) across three regularization strengths $C \in \{1, 10, 100\}$ into six members whose probabilities are averaged before the held-out fold is scored.

Algorithm 3 HER2 response: nested inner-CV summary-family selection (one outer fold)

- 1: split cohort $\rightarrow$ 17 test, 68 development (stratified by response)
 - 2: **for** each family $f \in \{\text{mean, median, P90}\}$ **do**
 - 3: **for** each inner stratified 5-fold split of the 68 development patients **do**
 - 4: standardize on the inner-train fold; rank genes by $|\text{corr}(\text{feature}, y)|$
 - 5: fit logistic regression on the top-5 features of family f ; predict the inner-validation fold
 - 6: record the pooled inner-fold AUROC for f
 - 7: $f^* \leftarrow \arg \max_f$ pooled inner-fold AUROC
 - 8: refit the top-5 model of family f^* on all 68 development patients
 - 9: predict $\hat{p}$ for the 17 test patients
-

HER2 response is the only task with a quantity left to select. The feature count is fixed at $K = 5$ as above, but each gene can be summarized across tiles three ways (mean, median, or the P90 hotspot), so an inner loop chooses that family without touching test patients: The full procedure runs over 50 repeats of stratified 5-fold splits; the 250 outer models yield out-of-fold probabilities that are averaged per patient before scoring, and 95% confidence intervals come from 2,000 bootstrap resamples of the patients. The P90 family was selected in 230 of the 250 outer folds, the mean in 14, and the median in 6. Gene-selection frequencies (how often each gene enters the top 5) are tallied across the same 250 models.

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